

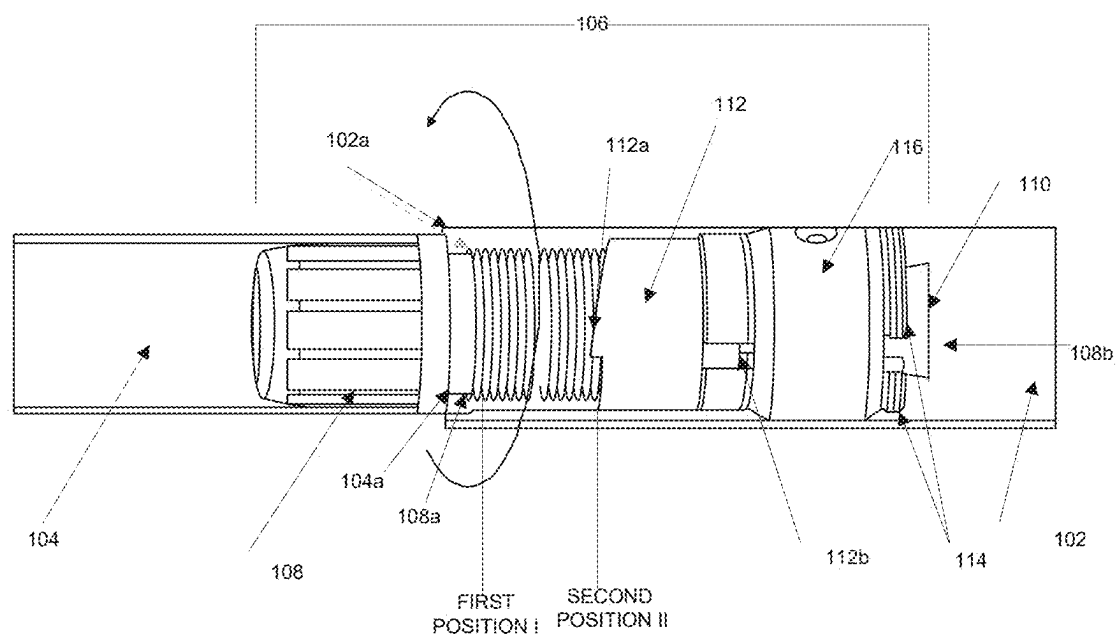


FIG. 1

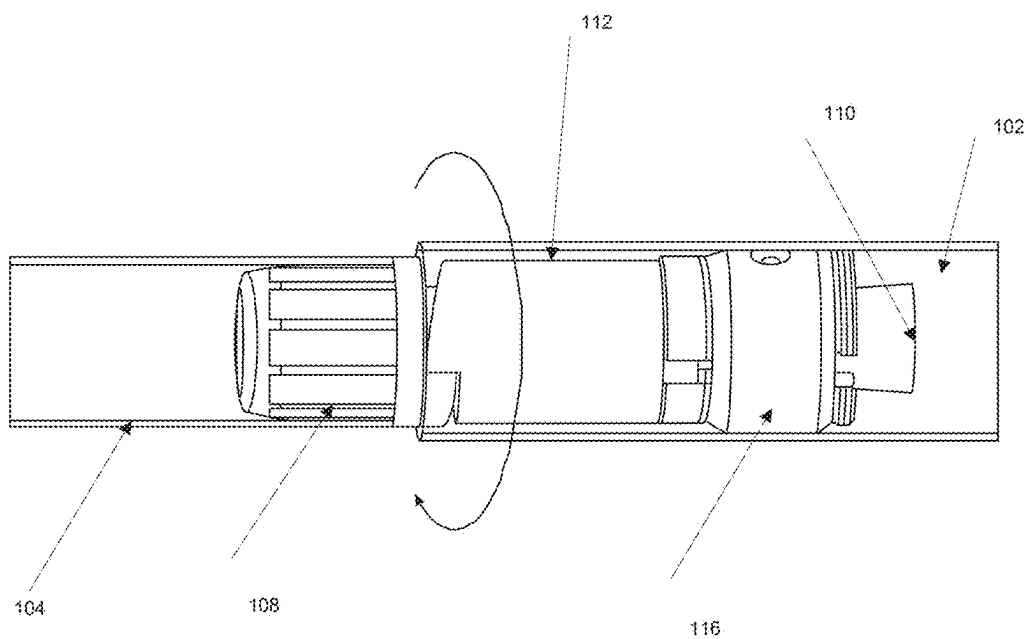


FIG. 2

EXTENDABLE ROD WITH A TWIST LOCK MECHANISM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority pursuant to 35 U.S.C. 119(a) to Indian Patent Application 202411012800, filed Feb. 22, 2024, which application is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to the technical field of adjustable rods for indoor decoration. More particularly, the present disclosure relates to an extendable rod with a twist lock mechanism.

BACKGROUND

[0003] Background description includes information that may be useful in understanding the present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

[0004] The conventional fixed curtain rods can be fixedly mounted against the walls using screws or nails. Further, specific tools are required to mount the rod. Unlike adjustable rods, fixed-length curtain rods have predetermined lengths, which can restrict their use to specific window or door sizes. Proper measurements are crucial to ensure a perfect fit of the rod against the supporting surfaces. The fixed curtain rods cannot be easily adjusted or extended in a situation when a person moves to a place with different dimensions of the windows or doors.

[0005] Further, adjustable length tension rods for use as curtain or shower rods are generally used as an alternative fixed rods that are permanently fixed to the support surfaces through the use of screws, bolts, or the like. The adjustable tensile rod provides for simpler installation and prevents permanent damage to the support surfaces upon removal. These tension rods typically include a first shaft that telescopically receives a second shaft, wherein the first and second shafts house along a threaded stud. The user may expand the tension rod for installation at the desired location. Once the desired length is reached, i.e. the end faces of the tension rod contact the walls or other support surfaces, the first and second shafts are rotated with respect to each other until a sufficient force is applied against the support surfaces to maintain the tension rod in compression and in place.

[0006] More recently, tension rods have been introduced that allow the user to easily expand the tension rod by pulling the first and second shafts in opposing directions. However, the tension rod must be pulled to a length that is larger than the distance between the support surfaces. Tension is then applied by compressing a spring within the tension rod until the desired length is achieved. Once in place, the force of the compressed spring provides the tension for holding the tension rod in place between the support surfaces. However, the spring can be difficult to compress, and positioning of the tension rod may scuff or damage the support surfaces as the rod is put into place. Further, if not properly locked into place, the spring may release and cause injury to the user.

[0007] There is, therefore, a need of an extendable rod with a twist lock mechanism that is easy to expand to the desired length, but does not require spring compression or the use of other hazardous mechanisms for providing the necessary tension between the support surfaces.

OBJECTS OF THE PRESENT DISCLOSURE

[0008] Some of the objects of the present disclosure, which at least one embodiment herein satisfies are as listed herein below.

[0009] An object of the present disclosure is to provide an extendable rod with a twist lock mechanism.

[0010] Another object of the present disclosure is to provide an extendable rod having better load bearing capacity.

[0011] Yet another object of the present disclosure is to eliminate the use of adhesive to mount the rods on walls/supporting surfaces.

[0012] Yet another object of the present disclosure is to provide an extendable rod with a twist lock mechanism that does not require drilling any holes in the wall to mount the extendable rods.

[0013] Yet another object of the present disclosure is to provide an extendable rod with a twist lock mechanism that does not require any special tools to adjust length of the extendable rod.

[0014] Yet another object of the present disclosure is to provide an extendable rod that is easy to install.

[0015] Yet another object of the present disclosure is to easily remove and/or replace curtains without dismantling the extendable rods.

SUMMARY

[0016] The present disclosure relates to adjustable rods for indoor decoration. More particularly the present disclosure relates to an extendable rod with a twist lock mechanism.

[0017] In aspect of the present disclosure pertains to an extendable rod with a twist lock mechanism. The extendable rod comprises a first tube and a second tube configured to be coaxially disposed within the first tube via an open end of the first tube. The extendable rod comprises a locking mechanism configured between the first tube and the second tube. The locking mechanism comprises a stud that comprises external threads at a first end. The stud is configured to be at least partially fitted into an open end of the second tube. The locking mechanism comprises a tapered mandrel configured at a second end of the stud which extends within the first tube upon coaxially disposing the second tube within the first tube. The tapered mandrel has a substantially frustoconical profile having diameter increasing in an axial direction moving away from the second tube. Further, the locking mechanism comprises a bush configured to be coaxially and rotationally disposed between an outer surface of the stud and an inner wall of the first tube at the open end of the first tube. A first end of the bush comprises internal threads that is adapted to engage with the external threads of the stud. Furthermore, the locking mechanism comprises at least two split bushes attached to or configured at a second end of the bush. The at least two split bushes remains coaxially disposed over an outer surface of the tapered mandrel. Moreover, the locking mechanism comprises a rubber sleeve coaxially disposed between an outer surface of the at least two split bushes and the inner wall of the first

tube. The locking mechanism is configured between the first tube and the second tube such that rotation of the second tube in a predefined direction with respect to the first tube causes the second tube to move from a first position to a second position away from the first tube and further restrict relative movement between the first tube and the second tube at the second position.

[0018] In an aspect, rotation of the first tube or the second tube in a clockwise direction may cause the second tube to move towards the second position and further enable movement of the bush away from the second tube. The movement of the bush away from the second tube results in sliding of the at least two split bushes over the tapered mandrel and correspondingly expanding the rubber sleeve.

[0019] In an aspect, the expansion of the rubber sleeve may create a frictional contact between the stud and the inner wall of the first tube to lock the second tube at the second position.

[0020] In an aspect, the clockwise rotation of the second tube or the first tube may move the second tube to the second position, thereby increasing the length of the rod and corresponding enable fitting of the rod against two opposite support surfaces.

[0021] In an aspect, rotation of the first tube or the second tube in a counter-clockwise direction may cause the second tube to move towards the first position and further enable movement of the bush away from the second tube. The movement of the bush away from the second tube results in sliding of the at least two split bushes away from the tapered mandrel and correspondingly contracting the rubber sleeve.

[0022] In an aspect, the bush may be configured to prevent anti-locking of the second tube and the first tube when rotated in the counter-clockwise direction.

[0023] In an aspect, the second tube and the first tube may be made of material selected from the group consisting of: cold-rolled steel, brass, high-grade steel, iron, aluminium or polymeric material.

[0024] In another aspect, the present disclosure pertains to a locking mechanism for an extendable tension rod comprising a first tube and a second tube coaxially disposed within the first tube. The locking mechanism comprises a stud that comprises external threads at a first end. The stud is configured to be at least partially fitted into an open end of the second tube. The locking mechanism comprises a tapered mandrel configured at a second end of the stud which extends within the first tube upon coaxially disposing the second tube within the first tube. The tapered mandrel has a substantially frustoconical profile having diameter increasing in an axial direction moving away from the second tube. Further, the locking mechanism comprises a bush configured to be coaxially and rotationally disposed between an outer surface of the stud and an inner wall of the first tube at the open end of the first tube. A first end of the bush comprises internal threads that is adapted to engage with the external threads of the stud. Furthermore, the locking mechanism comprises at least two split bushes attached to or configured at a second end of the bush, such that the at least two split bushes remains coaxially disposed over an outer surface of the tapered mandrel. Moreover, the locking mechanism comprises a rubber sleeve coaxially disposed between an outer surface of the at least two split bushes and the inner wall of the first tube. The locking mechanism is configured between the first tube and the second tube such that rotation of the second tube in a

predefined direction with respect to the first tube causes the second tube to move from a first position to a second position away from the first tube and further restrict relative movement between the first tube and the second tube at the second position.

[0025] In another aspect, the tapered mandrel may be coupled to the stud at the second end using a thread connection.

[0026] In another aspect, the at least two split bushes may comprise an expansion anchor integrally formed with the bush.

[0027] Various objects, features, aspects, and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like components.

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] The following drawings form part of the present specification and are included to further illustrate aspects of the present disclosure. The disclosure may be better understood by reference to the drawings in combination with the detailed description of the specific embodiments presented herein.

[0029] FIG. 1 illustrates a schematic diagram of an extendable tensile rod with a twist lock mechanism in a locked position, in accordance with an embodiment of the present disclosure.

[0030] FIG. 2 illustrates a schematic diagram of an extendable tensile rod with a twist lock mechanism in an unlocked position, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0031] The following is a detailed description of embodiments of the disclosure depicted in the accompanying drawings. The embodiments are in such detail as to clearly communicate the disclosure. If the specification states a component or feature “may”, “can”, “could”, or “might” be included or have a characteristic, that particular component or feature is not required to be included or have the characteristic.

[0032] As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

[0033] According to an embodiment, the present disclosure relates to an extendable rod with a twist lock mechanism that is easy to use, economical and have better weight bearing capabilities. The proposed extendable rod can be used as curtain rod, shower rod or the like, does not require any adhesive to mount the rod against opposite supporting surfaces.

[0034] Further, the proposed locking mechanism for the extendable rod can expand the rod both in length and in diameter which results in producing desired force to lock and hold the rod against the opposite supporting surfaces. Further, the locking mechanism is designed in such a manner that it does not permit locking in reverse direction.

[0035] The present invention is useful for supporting shower curtains, window curtains, and door curtains or the

like, including in telescoping tube assemblies. Traditional fixed curtain rods can be fixedly mounted against the walls using screws or nails. Further, specific tools are required to mount the rod. Unlike adjustable rods, fixed-length curtain rods have predetermined lengths, which can restrict their use to specific window sizes. Proper measurements are crucial to ensure a perfect fit of the rod against the supporting surfaces. The fixed curtain rods cannot be easily adjusted or extended in a situation when a person moves to a place with different dimension of the window or door. The proposed extendable and adjustable rod overcomes the abovementioned problems associated with the tradition rods. The unique design of the locking mechanism implemented in the extendable rod provides advantageous features such as versatile length options, easy installation without need of drilling or additional tools, cost-effective and reliable rod.

[0036] In an embodiment, the extendable rod can include a first tube and a second tube. The extendable rod may be configured to be coaxially disposed within the first tube via an open end of the first tube.

[0037] In an embodiment, the locking mechanism may be configured between the first tube and the second tube. The locking mechanism is configured between the first tube and the second tube such that rotation of the second tube in a predefined direction with respect to the first tube causes the second tube to move from a first position to a second position away from the first tube and further restrict relative movement between the first tube and the second tube at the second position. In an exemplary embodiment, the second tube and the first tube can be made of material selected from the group consisting of cold-rolled steel, brass, high-grade steel, iron, aluminium or polymeric material.

[0038] In an embodiment, the stud may include external threads at a first end. The stud may be configured to be at least partially fitted into an open end of the second tube.

[0039] In an embodiment, the tapered mandrel may be configured at a second end of the stud. The tapered mandrel may extend within the first tube upon coaxially disposing the second tube within the first tube. The tapered mandrel may have a substantially frustoconical profile. The frustoconical profile may have a diameter increasing in an axial direction moving away from the second tube.

[0040] In an embodiment, the bush may be configured to be coaxially and rotationally disposed. The stud may be configured between an outer surface of the stud and an inner wall of the first tube at the open end of the first tube. The bush may include internal threads at first end. The internal threads may be adapted to engage with the external threads of the stud. The movement of the bush away from the second tube results in sliding of the at least two split bushes away from the tapered mandrel and correspondingly contracting the rubber sleeve. The bush can be configured to prevent anti-locking of the second tube and the first tube when rotated in the counter-clockwise direction.

[0041] In an embodiment, the at least two split bushes may be attached to or configured at a second end of the bush. The at least two split bushes may remain coaxially disposed over an outer surface of the tapered mandrel.

[0042] In an embodiment, the rubber sleeve may coaxially disposed between an outer surface of the at least two split bushes and the inner wall of the first tube. The expansion of the rubber sleeve can create a frictional contact between the stud and the inner wall of the first tube to lock the second tube at the second position.

[0043] In an embodiment, the clockwise rotation of the second tube or the first tube can move the second tube to the second position, thereby increasing the length of the rod and corresponding enable fitting of the rod against two opposite support surfaces. The rotation of the first tube or the second tube in a counter-clockwise direction can cause the second tube to move towards the first position and further enable movement of the bush away from the second tube.

[0044] In an embodiment, the locking mechanism equipped with the twist lock independently as the system can be configured on any other extendable rod besides curtain rods.

[0045] Referring to FIG. 1, the extendable rod **100** comprises a first tube **102** and a second tube **104** configured to be coaxially disposed within the first tube **102** through an open end **102a** of the first tube **102**. The extendable rod **100** comprises a locking mechanism **106** configured between the first tube **102** and the second tube **104**. The first tube **102** can also be referred herein as “outer tube” and the second tube **104** can be interchangeably used as “inner tube”.

[0046] In an embodiment, the locking mechanism **106** comprises a stud **108** that comprises external threads at a first end **108a**. The stud **108** is configured to be at least partially fitted into an open end **104a** of the second tube **104**. The locking mechanism **106** comprises a tapered mandrel **110** configured at a second end **108b** of the stud **108** which extends within the first tube **102** upon coaxially disposing the second tube **104** within the first tube **102**. The tapered mandrel **110** has a substantially frustoconical profile having diameter increasing in an axial direction moving away from the second tube **104**. Alternatively, the tapered mandrel **110** can be coupled to the stud **108** at the second end using a thread connection.

[0047] Further, the locking mechanism **106** comprises a bush configured to be coaxially and rotationally disposed between an outer surface of the stud **108** and an inner wall of the first tube **102** at the open end of the first tube **102**. A first end **112a** of the bush **112** comprises internal threads that is adapted to engage with the external threads of the stud **108**. Furthermore, the locking mechanism **106** comprises at least two split bushes **114** attached to or configured at a second end **112b** of the bush **112** in a manner that the at least two split bushes **114** remains coaxially disposed over an outer surface of the tapered mandrel **110**.

[0048] Additionally, the locking mechanism **106** comprises a rubber sleeve **116** coaxially disposed between an outer surface of the at least two split bushes **114** and the inner wall of the first tube **102**. The rubber sleeve **116** is preferably made from polymeric material such as polycarbonate, polystyrene, polypropylene, Acrylonitrile butadiene styrene (ABS), Styrene acrylonitrile resin (SAN), polyurethane, Polyvinyl chloride (PVC), rubber, or the like.

[0049] The locking mechanism **106** is configured between the first tube **102** and the second tube **104** such that rotation of the second tube **104** in a predefined direction with respect to the first tube **102** causes the second tube **104** to move from a first position I to a second position II away from the first tube **102** and further restrict relative movement between the first tube **102** and the second tube **104** at the second position II.

[0050] Referring to FIG. 1, an extendable rod **100** with a twist lock mechanism **106** in a locked position is disclosed. A rotation of the first tube **102** or the second tube **104** in a clockwise direction can cause the second tube **104** to move

towards the second position and further enable movement of the bush 112 away from the second tube 104, resulting in sliding of the at least two split bushes 114 over the tapered mandrel 110 and correspondingly expanding the rubber sleeve 116. The expansion of the rubber sleeve 116 can create a frictional contact between the stud 108 and the inner wall of the first tube 102 to lock the second tube 104 at the second position II.

[0051] Referring to FIG. 2, the extendable rod 100 with a twist lock mechanism 106 in an unlocked position is disclosed. The rotation of the first tube 102 or the second tube 104 in a counter-clockwise direction causes the second tube 104 to move towards the first position I and further enable movement of the bush 112 away from the second tube 104, resulting in sliding of the at least two split bushes 114 away from the tapered mandrel 110 and correspondingly contracting the rubber sleeve 116. The bush 112 can be configured to prevent anti-locking of the second tube 104 and the first tube 102 when rotated in the counter-clockwise direction. The stud 108 can be selected from but not limited to flange studs, flange studs with chamfered ends, tap end studs and double end studs. In an alternate embodiment, bolt with flange can also be used in place of the stud 108. The at least two split bushes 114 can include an expansion anchor integrally formed with the bush 112. The second tube 104 and the first tube 102 can be made of material selected from the group consisting of but not limited to cold-rolled steel, brass, high-grade steel, iron, aluminium or polymeric material.

[0052] Those skilled in the art would appreciate that the proposed unique design of components of the twist-locking mechanism 106 of the proposed extendable rod 100, which are designed or manufactured in such a way that the bush prevents anti-locking of the second tube and the first tube when rotated in the counter-clockwise direction. Further, the extendable rod 100 can be locked at desirable length simply by twisting either the first tube 102 or the second tube 104. Moreover, the extendable rod can be easily mounted against the supporting surfaces without need of any specific tools or drilling.

[0053] Thus, the present disclosure overcomes the drawbacks, shortcomings, and limitations associated with the existing extendable rods and associated locking mechanisms, by providing an improved and simple twist-locking mechanism for locking the extendable rod at desirable length, having advantageous features included with unique design, including, for example versatile length options, easy installation without need of drilling or additional tools, cost-effective, reliable design, better serviceability and easy assembly or disassembly of components of the extendable rod.

[0054] Moreover, in interpreting the specification, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced. Where the specification claims refer to at least one of something selected from the group consisting of A, B, C . . . and N, the text should be interpreted as requiring only one element from the group, not A plus N, or B plus N, etc.

[0055] While the foregoing describes various embodiments of the invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof. The scope of the invention is determined by the claims that follow. The invention is not limited to the described embodiments, versions or examples, which are included to enable a person having ordinary skill in the art to make and use the invention when combined with information and knowledge available to the person having ordinary skill in the art.

ADVANTAGES OF THE PRESENT DISCLOSURE

[0056] The present invention provides an extendable rod with a twist lock mechanism.

[0057] The present invention provides an extendable rod having better load bearing capacity.

[0058] The present invention provides an extendable rod that eliminate the use of adhesive to mount the rod on walls.

[0059] The present invention provides an extendable rod with a twist lock mechanism that does not require reverse locking.

[0060] The present invention provides an extendable rod with a twist lock mechanism that does not require drilling any holes in the wall to mount the extendable rod.

[0061] The present invention provides an extendable rod with a twist lock mechanism that does not require any special tools to adjust length of the extendable rod.

I claim:

1. An extendable rod comprising:

a first tube;

a second tube configured to be coaxially disposed within the first tube via an open end of the first tube; and

a locking mechanism configured between the first tube and the second tube, wherein the locking mechanism comprises:

a stud that comprises external threads at a first end that is configured to be at least partially fitted into an open end of the second tube, and wherein a tapered mandrel is configured at a second end of the stud which extends within the first tube upon coaxially disposing the second tube within the first tube, wherein the tapered mandrel has a substantially frustoconical profile having diameter increasing in an axial direction moving away from the second tube;

a bush configured to be coaxially and rotationally disposed between an outer surface of the stud and an inner wall of the first tube at the open end of the first tube, wherein a first end of the bush comprises internal threads that is adapted to engage with the external threads of the stud;

at least two split bushes attached to or configured at a second end of the bush, such that the at least two split bushes remains coaxially disposed over an outer surface of the tapered mandrel; and

a rubber sleeve coaxially disposed between an outer surface of the at least two split bushes and the inner wall of the first tube;

wherein the locking mechanism is configured between the first tube and the second tube such that rotation of the second tube in a predefined direction with respect to the first tube causes the second tube to move from a first position to a second position away

from the first tube and further restrict relative movement between the first tube and the second tube at the second position.

2. The extendable rod as claimed in claim 1, wherein rotation of the first tube or the second tube in a clockwise direction causes the second tube to move towards the second position and further enable movement of the bush away from the second tube, resulting in sliding of the at least two split bushes over the tapered mandrel and correspondingly expanding the rubber sleeve.

3. The extendable rod as claimed in claim 2, wherein the expansion of the rubber sleeve creates a frictional contact between the stud and the inner wall of the first tube to lock the second tube at the second position.

4. The extendable rod as claimed in claim 2, wherein the clockwise rotation of the second tube or the first tube moves the second tube to the second position, thereby increasing the length of the rod and corresponding enable fitting of the rod against two opposite support surfaces.

5. The extendable rod as claimed in claim 1, wherein rotation of the first tube or the second tube in a counter-clockwise direction causes the second tube to move towards the first position and further enable movement of the bush away from the second tube, resulting in sliding of the at least two split bushes away from the tapered mandrel and correspondingly contracting the rubber sleeve.

6. The extendable rod as claimed in claim 5, wherein the bush is configured to prevent anti-locking of the second tube and the first tube when rotated in the counter-clockwise direction.

7. The extendable rod as claimed in claim 1, wherein the second tube and the first tube are made of material selected from the group consisting of: cold-rolled steel, brass, high-grade steel, iron, aluminium or polymeric material.

8. A locking mechanism for an extendable rod comprising a first tube and a second tube coaxially disposed within the first tube, the locking mechanism comprising:

a stud that comprises external threads at a first end that is configured to be at least partially fitted into an open end of the second tube, and wherein a tapered mandrel is configured at a second end of the stud which extends within the first tube upon coaxially disposing the second tube within the first tube, wherein the tapered mandrel has a substantially frustoconical profile having diameter increasing in an axial direction moving away from the second tube;

a bush configured to be coaxially and rotationally disposed between an outer surface of the stud and an inner wall of the first tube at the open end of the first tube, wherein a first end of the bush comprises internal threads that is adapted to engage with the external threads of the stud;

at least two split bushes attached to or configured at a second end of the bush, such that the at least two split bushes remains coaxially disposed over an outer surface of the tapered mandrel; and

a rubber sleeve coaxially disposed between an outer surface of the at least two split bushes and the inner wall of the first tube;

wherein the locking mechanism is configured between the first tube and the second tube such that rotation of the second tube in a predefined direction with respect to the first tube causes the second tube to move from a first position to a second position away from the first tube and further restrict relative movement between the first tube and the second tube at the second position.

9. The locking mechanism as claimed in claim 8, wherein the tapered mandrel is coupled to the stud at the second end using a thread connection.

10. The locking mechanism as claimed in claim 8, wherein the at least two split bushes comprises an expansion anchor integrally formed with the bush.

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